Sommerfeld, “I am now working exclusively on the gravity problem…. [O]ne thing is certain—that never in my life have I tormented myself anything like this. . . . Compared to this problem the original [i.e., special] relativity theory is child’s play.”3

He appears to have made the next key breakthrough, a simple yet subtle consequence of applying special relativity to the link between gravity and accelerated motion, in 1912. To understand this step in Einstein’s reasoning it is easiest to focus, as apparently he did, on a particular example of accelerated motion.4 Recall that an object is accelerating if either the speed or the direction of its motion changes. For simplicity we will focus on accelerated motion in which only the direction of our object’s motion changes while its speed stays fixed. Specifically, we consider motion in a circle such as what one experiences on the Tornado ride in an amusement park. In case you have never tested the stability of your constitution on this ride, you stand with your back against the inside of a circular Plexiglas structure that spins at a high speed. Like all accelerated motion, you can feel this motion—you feel your body being pulled radially away from the ride’s center and you feel the circular wall of Plexiglas pressing on your back, keeping you moving in a circle. (In fact, although not relevant for the present discussion, the spinning motion “pins” your body to the Plexiglas with such a force that when the ledge on which you are standing drops away you do not slip downward.) If the ride is extremely smooth and you close your eyes, the pressure of the ride on your back—like the support of a bed—can almost make you feel that you are lying down. The “almost” comes from the fact that you still feel ordinary “vertical” gravity, so your brain cannot be fully fooled. But if you were to ride the Tornado in outer space, and if it were to spin at just the right rate, it would feel just like lying in a stationary bed on earth. Moreover, were you to “get up” and walk along the interior of the spinning Plexiglas, your feet would press against it just as they do against an earthbound floor. In fact, space stations are designed to spin in this manner to create an artificial feeling of gravity in outer space.

Having used the accelerated motion of the spinning Tornado to imitate gravity, we can now follow Einstein and set out to see how space and time appear to someone on the ride. His reasoning, adapted